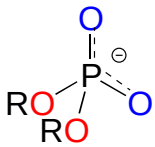
major resonance contributors

minor resonance contributors
(separation of charge)



non-bridging bonds: significant double bond character

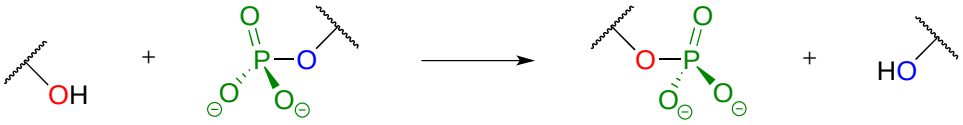
bridging bonds: little double-bond character

Exercise 9.4: Draw all of the resonance structures showing the delocalization of charge on a (fully deprotonated) organic monophosphate. If a 'bond order' of 1.0 is a single bond, and a bond order of 2.0 is a double bond, what is the approximate bond order of bridging and non-bridging P-O bonds?

Throughout this book, phosphate groups will often be drawn without attempting to show tetrahedral geometry, and π bonds and negative charges will usually be shown localized to a single oxygen. This is done for the sake of simplification - however it is important always to remember that the phosphate group is really tetrahedral, the negative charges are delocalized over the non-bridging oxygens, and that there is some degree of protonation at physiological pH (with the exception of the phosphate diester group).

Section 9.2: Phosphate transfer reactions - an overview

In a phosphate transfer reaction, a phosphate group is transferred from a **phosphate group donor** molecule to a **phosphate group acceptor** molecule:



phosphate acceptor

phosphate donor

A very important aspect of biological phosphate transfer reactions is that the electrophilicity of the phosphorus atom is usually enhanced by the Lewis acid (electron-accepting) effect of one or more magnesium ions. Phosphate transfer enzymes generally